Closed Subschemes

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Exercise 1.6

We say that a topological space X is **quasi-compact** if from any open covering $\{U_i\}_i$ of X, we can extract a finite subcovering.

(a) Show that any closed subspace of a quasi-compact space is quasi-compact.

(b) Show that $\operatorname{Spec} A$ is quasi-compact for any ring A.

Solution.

(a) Closed subspaces of quasi-compact spaces are quasi-compact.

Let X be a quasi-compact space and let $Z \subseteq X$ be a closed subspace. Let $\{V_i\}_i$ be an open covering of Z. Since Z has the subspace topology, each V_i can be written as $V_i = U_i \cap Z$ for some open $U_i \subseteq X$.

Since Z is closed, $X \setminus Z$ is open. Consider the open covering of X given by:

$$\{U_i\}_i \cup \{X \setminus Z\}.$$

Since X is quasi-compact, we can extract a finite subcovering. There are two cases:

1. **If $X \setminus Z$ is not used** in the finite subcovering, then we have $\{U_{i_1}, \dots, U_{i_n}\}$ covering X. Since $Z \subseteq X$, these sets also cover Z, and thus $\{V_{i_1}, \dots, V_{i_n}\}$ where $V_{i_k} = U_{i_k} \cap Z$ cover Z.

2. **If $X \setminus Z$ is used** in the finite subcovering, say $\{U_{i_1}, \dots, U_{i_n}, X \setminus Z\}$, then removing $X \setminus Z$ still covers Z (since $X \setminus Z$ doesn't intersect Z). Thus $\{V_{i_1}, \dots, V_{i_n}\}$ covers Z.

In either case, we've extracted a finite subcovering of $\{V_i\}_i$ from the original covering of Z. Hence Z is quasi-compact.

(b) $\operatorname{Spec} A$ **is quasi-compact.**

Recall that the **standard open subsets** (also called **principal open subsets**) of $\operatorname{Spec} A$ are sets of the form:

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}$$

for $f \in A$. These form a basis for the Zariski topology on $\operatorname{Spec} A$.

Let $\{U_i\}_i$ be an open covering of $\operatorname{Spec} A$. Each U_i can be written as a union of standard opens. By refining the covering if necessary, we may assume that each U_i is itself a standard open $D(f_i)$ for some $f_i \in A$.

Thus we have a covering:

$$\operatorname{Spec} A = \bigcup_{i \in I} D(f_i).$$

We claim that this implies the ideal generated by $\{f_i\}_{i \in I}$ is the whole ring A. Suppose not. Then the ideal $(\{f_i\}_{i \in I})$ is contained in some maximal ideal $\mathfrak{m} \subset A$. This means $f_i \in \mathfrak{m}$ for all $i \in I$, so $\mathfrak{m} \notin D(f_i)$ for any i. But then $\mathfrak{m} \notin \bigcup_i D(f_i)$, contradicting that the $D(f_i)$ cover $\operatorname{Spec} A$.

Therefore, $(\{f_i\}_{i \in I}) = A$, and there exists a finite subset $\{f_{i_1}, \dots, f_{i_n}\}$ such that:

$$(f_{i_1}, \dots, f_{i_n}) = A.$$

This is because 1 can be expressed as a finite linear combination of the f_i:

$$1 = \sum_{k=1}^{n} a_k f_{i_k}$$

for some $a_k \in A$ and finitely many f_{i_k}.

Key observation: If $(f_1, \dots, f_n) = A$, then $D(f_1) \cup \cdots \cup D(f_n) = \operatorname{Spec} A$.

Proof. For any prime ideal $\mathfrak{p} \in \operatorname{Spec} A$, if $\mathfrak{p}$ were not in any $D(f_k)$, then $f_k \in \mathfrak{p}$ for all $k = 1, \dots, n$. But then:

$$1 = \sum_{k=1}^{n} a_k f_k \in \mathfrak{p},$$

which is impossible since $\mathfrak{p} \neq A$.

Therefore $\{D(f_{i_1}), \dots, D(f_{i_n})\}$ is a finite subcovering of $\{D(f_i)\}_{i \in I}$, and $\operatorname{Spec} A$ is quasi-compact.

Intuition. The proof uses a crucial algebraic fact: if an ideal equals the whole ring, then 1 can be written as a *finite* linear combination of generators. This finiteness property of rings is what guarantees the quasi-compactness of $\operatorname{Spec} A$.

Lemma 2.9

Let F be a sheaf on X. Let $s, t \in F(X)$ be sections such that $s_x = t_x$ for every $x \in X$. Then $s = t$.

Proof. We may assume that $t = 0$. For every $x \in X$, there exists an open neighborhood U_x of x such that $s|_{U_x} = 0$, because $s_x = 0$. As the U_x cover X when x varies, we indeed have $s = 0$.

Proposition 2.12

Let $\alpha : F \to G$ be a morphism of sheaves on X. Then α is an isomorphism if and only if α_x is an isomorphism for every $x \in X$.

Proof. Suppose that α_x is an isomorphism for every $x \in X$. Let us show that α is an isomorphism (the converse is trivial). Let $s \in F(U)$ be a section. If $\alpha(U)(s) = 0$, then for every $x \in U$, we have $\alpha_x(s_x) = (\alpha(U)(s))_x = 0$. It follows that $s_x = 0$ for every $x \in U$. Hence $s = 0$ (Lemma 2.9) and α is injective. Next, let $t \in G(U)$. Then we can find a covering of U by open sets U_i and sections $s_i \in F(U_i)$ such that $\alpha(U_i)(s_i) = t|_{U_i}$. As we have just seen that α is injective, s_i and s_j coincide on $U_i \cap U_j$. The s_i therefore glue to a section $s \in F(U)$ (i.e., $s|_{U_i} = s_i$). By construction, $\alpha(U)(s)$ and t coincide on every U_i, and are therefore equal. This proves that $\alpha(U)$ is surjective.

Proposition 2.18

A sequence of sheaves $F \to G \to H$ is exact if and only if

$$F_x \to G_x \to H_x$$

is an exact sequence of groups for every $x \in X$.

Topological Immersions

Before defining closed immersions of ringed spaces, we recall the purely topological notions.

A **topological open immersion** is a continuous map $f : X \to Y$ such that f is a homeomorphism of X onto an open subset $f(X) \subseteq Y$. Equivalently, f identifies X with an open subset of Y.

A **topological closed immersion** is a continuous map $f : X \to Y$ such that f is a homeomorphism of X onto a closed subset $f(X) \subseteq Y$. Equivalently, f identifies X with a closed subset of Y.

Examples. - The inclusion $\{0\} \hookrightarrow \mathbb{R}$ is a closed immersion (a point is closed in $\mathbb{R}$). - The inclusion $(0,1) \hookrightarrow \mathbb{R}$ is an open immersion but not a closed immersion. - The inclusion $[0,1] \hookrightarrow \mathbb{R}$ is neither open nor closed as a map of topological spaces (the image is not open, and while $[0,1]$ is closed, the map is not a homeomorphism onto its image with the subspace topology).

Definition 2.22

We say that a morphism $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is an open immersion (resp. closed immersion) if f is a topological open immersion (resp. closed immersion) and if $f_x^\#$ is an isomorphism (resp. if $f_x^\#$ is surjective) for every $x \in X$.

Remark. The different conditions on $f_x^\#$ reflect the algebraic nature of these constructions.

- For an **open immersion**, X is identified with an open subset $V \subseteq Y$. The sheaf $\mathcal{O}_X$ is literally the restriction of $\mathcal{O}_Y$ to V. At the level of stalks, functions on Y near $x \in V$ are the same as functions on X near x—we're just looking at the same data. Hence $f_x^\#$ must be an isomorphism. This is analogous to **localization** in algebra: if $S^{-1}A$ is a localization of A, the natural map $A \to S^{-1}A$ becomes an isomorphism after localizing.
- For a **closed immersion**, X is identified with a closed subset $Z \subseteq Y$. The sheaf $\mathcal{O}_X$ is a "quotient" of $\mathcal{O}_Y$ by functions that vanish on Z. At the level of stalks, we have a surjection $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ whose kernel consists of germs of functions vanishing at x. This is not injective unless the closed subset is the whole space. This is analogous to **quotient by an ideal** in algebra: the map $A \to A/J$ is surjective but not injective (unless $J = 0$).

In summary:

$$\text{open immersion} \longleftrightarrow \text{localization} \quad \text{(injective)}$$

$$\text{closed immersion} \longleftrightarrow \text{quotient} \quad \text{(surjective)}$$

Let V be an open subset of Y. Then $(V, \mathcal{O}_Y|_V)$ is clearly a ringed topological space. In all that follows, an open subset V will always be endowed with this structure induced by $\mathcal{O}_Y$. It is clear that $(f, f^\#)$ is an open immersion if and only if there exists an open subset V of Y such that $(f, f^\#)$ induces an isomorphism from $(X, \mathcal{O}_X)$ onto $(V, \mathcal{O}_Y|_V)$ (see Proposition 2.12).

Indeed, the topological condition implies $f(X) = V$ is open, and the condition that $f_x^\#$ is an isomorphism for every $x \in X$ means precisely that the induced morphism $(f, f^\#) : (X, \mathcal{O}_X) \to (V, \mathcal{O}_Y|_V)$ is an isomorphism of ringed spaces by Proposition 2.12.

In what follows, we will give a characterization of the closed immersions. Let $(X, \mathcal{O}_X)$ be a ringed topological space. Let $\mathcal{J}$ be a sheaf of ideals of $\mathcal{O}_X$ (i.e., $\mathcal{J}(U)$ is an ideal of $\mathcal{O}_X(U)$ for every open subset U). Let

$$V(\mathcal{J}) := \{x \in X \mid \mathcal{J}_x \neq \mathcal{O}_{X,x}\}.$$

Notation. The notation $V(\mathcal{J})$ generalizes the classical algebraic geometry notion of the "variety" or "zero set" of an ideal. In classical algebraic geometry, for an ideal $I \subseteq k[x_1, \ldots, x_n]$, one defines

$$V(I) = \{x \in \mathbb{A}^n \mid f(x) = 0 \text{ for all } f \in I\}.$$

Equivalently, $V(I)$ consists of points where the localization I_x is not equal to the whole ring $\mathcal{O}_{\mathbb{A}^n,x}$.

In the sheaf-theoretic setting, a point $x \in V(\mathcal{J})$ precisely when $\mathcal{J}_x \subsetneq \mathcal{O}_{X,x}$—that is, when the stalk of $\mathcal{J}$ at x is a *proper* ideal. This means there exist functions in $\mathcal{J}$ that "vanish" at x (more precisely, the germ of some section of $\mathcal{J}$ at x is not invertible).

Thus $V(\mathcal{J})$ is the set of points "where $\mathcal{J}$ defines a proper equation"—the closed subset cut out by the vanishing of functions in $\mathcal{J}$.

Lemma 2.23

Let X be a ringed topological space. Let $\mathcal{J}$ be a sheaf of ideals of $\mathcal{O}_X$. Let $j : V(\mathcal{J}) \to X$ denote the inclusion. Then $V(\mathcal{J})$ is a closed subset of X, $(V(\mathcal{J}), j^{-1}(\mathcal{O}_X/\mathcal{J}))$ is a ringed topological space, and we have a closed immersion $(j, j^{\#})$ of this space into $(X, \mathcal{O}_X)$, where $j^{\#}$ is the canonical surjection $\mathcal{O}_X \to \mathcal{O}_X/\mathcal{J} = j_*(j^{-1}(\mathcal{O}_X/\mathcal{J}))$.

Proof. If $x \in X \setminus V(\mathcal{J})$, there exist an open neighborhood $U \ni x$ and $f \in \mathcal{J}(U)$ such that $f_x = 1$. It follows that $f|_V = 1$ in an open neighborhood $V \subseteq U$ of x. We then have $V \subseteq X \setminus V(\mathcal{J})$. The latter is therefore open. For every $x \in V(\mathcal{J})$, $(j^{-1}(\mathcal{O}_X/\mathcal{J}))_x = (\mathcal{O}_X/\mathcal{J})_x = \mathcal{O}_{X,x}/\mathcal{J}_x$ is a local ring. The rest follows immediately.

In the rest of this book, given a sheaf of ideals $\mathcal{J}$, we will always endow $V(\mathcal{J})$ with the structure described in this lemma.

Proposition 2.24

Let $f : Y \to X$ be a closed immersion of ringed topological spaces. Let Z be the ringed topological space $V(\mathcal{J})$ where $\mathcal{J} = \operatorname{Ker} f^{\#} \subseteq \mathcal{O}_X$. Then f factors into an isomorphism $Y \simeq Z$ followed by the canonical closed immersion $Z \to X$.

Proof. As $f(Y)$ is closed in X, we have

$$(f_*\mathcal{O}_Y)_x = \begin{cases} 0 & \text{if } x \notin f(Y) \\ \mathcal{O}_{Y,y} & \text{if } x = f(y). \end{cases}$$

Using the exact sequence $0 \to \mathcal{J} \to \mathcal{O}_X \to f_*\mathcal{O}_Y \to 0$ and Proposition 2.18, we obtain exactness on stalks:

$$0 \to \mathcal{J}_x \to \mathcal{O}_{X,x} \to (f_*\mathcal{O}_Y)_x \to 0.$$

- If $x \notin f(Y)$, then $(f_*\mathcal{O}_Y)_x = 0$. The map $\mathcal{O}_{X,x} \to 0$ is the zero map, so by exactness at $\mathcal{J}_x$, the map $\mathcal{J}_x \to \mathcal{O}_{X,x}$ is an isomorphism. Hence $\mathcal{J}_x = \mathcal{O}_{X,x}$.
- If $x \in f(Y)$, then $(f_*\mathcal{O}_Y)_x = \mathcal{O}_{Y,y} \neq 0$. The map $\mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$ is surjective (since f is a closed immersion), so its kernel $\mathcal{J}_x$ is a proper ideal of $\mathcal{O}_{X,x}$. Hence $\mathcal{J}_x \neq \mathcal{O}_{X,x}$.

Therefore $\mathcal{J}_x = \mathcal{O}_{X,x}$ if and only if $x \notin f(Y)$, which means $x \in V(\mathcal{J})$ if and only if $x \in f(Y)$. Hence $V(\mathcal{J}) = f(Y)$.

Now we construct the isomorphism $Y \simeq Z$. Let $j : Z \to X$ be the canonical injection. Since $Z = V(\mathcal{J}) = f(Y)$ as sets, we can define $g : Y \to Z$ to be the homeomorphism induced by f—that is, g is the bijection $f(Y) \to V(\mathcal{J})$ given by the identity map on underlying points. Then $f = j \circ g$ as continuous maps of topological spaces.

Step 1: Relating the sheaves. By construction of $Z = V(\mathcal{J})$, the space Z is endowed with the structure $(V(\mathcal{J}), j^{-1}(\mathcal{O}_X/\mathcal{J}))$ as defined in Lemma 2.23. In other words, $\mathcal{O}_Z = j^{-1}(\mathcal{O}_X/\mathcal{J})$. Thus:

$$j_*\mathcal{O}_Z = j_*j^{-1}(\mathcal{O}_X/\mathcal{J}).$$

But $\mathcal{O}_X/\mathcal{J} \simeq f_*\mathcal{O}_Y$ (this is exactly what the exact sequence $0 \to \mathcal{J} \to \mathcal{O}_X \to f_*\mathcal{O}_Y \to 0$ tells us). Therefore:

$$j_*\mathcal{O}_Z \simeq j_*j^{-1}(f_*\mathcal{O}_Y).$$

Step 2: Applying Exercise 2.6. We have $f = j \circ g$, so by Exercise 2.6(b):

$$f_*\mathcal{O}_Y = (j \circ g)_*\mathcal{O}_Y = j_*(g_*\mathcal{O}_Y).$$

Substituting this into our expression:

$$j_*\mathcal{O}_Z \simeq j_*j^{-1}j_*(g_*\mathcal{O}_Y).$$

Step 3: Applying Exercise 2.13. Since j is a closed immersion, Exercise 2.13(a) tells us that the canonical morphism $G \to j_* j^{-1} G$ is surjective for any sheaf G. More importantly, Exercise 2.13(b) gives us an adjunction. But here we need something more specific: for an open immersion, $j^{-1} j_* F \to F$ is an isomorphism (Exercise 2.13(a)). However, j is a **closed** immersion, not open.

Instead, we use the fact that the unit $G \to j_* j^{-1} G$ is the counit of the adjunction applied to $j^{-1} G$. Since j is the inclusion of a closed subset, we can check this equality on stalks. For $x \in Z$:

$$(j_* j^{-1} j_* (g_* \mathcal{O}_Y))_x = (j^{-1} j_* (g_* \mathcal{O}_Y))_x = (j_* (g_* \mathcal{O}_Y))_x = (g_* \mathcal{O}_Y)_x,$$

where we used that j^{-1} doesn't change stalks for points in the image of j.

Step 4: The key isomorphism. From the adjunction of Exercise 2.13(b), we have a canonical isomorphism:

$$\mathcal{O}_Z = j^{-1} j_* \mathcal{O}_Z \simeq j^{-1} j_* (g_* \mathcal{O}_Y).$$

But now $j^{-1} j_* (g_* \mathcal{O}_Y) \simeq g_* \mathcal{O}_Y$ because $g_* \mathcal{O}_Y$ is already a sheaf on Z (by definition of the pushforward), and $j^{-1} j_*$ acts as the identity on sheaves supported on Z. More formally, the counit map $j^{-1} j^* F \to F$ is an isomorphism when j is a topological embedding (which it is, as a closed immersion).

Therefore:

$$\mathcal{O}_Z \simeq g_* \mathcal{O}_Y.$$

Conclusion. The map $g : Y \to Z$ is a homeomorphism by construction, and we've shown $\mathcal{O}_Z \simeq g_* \mathcal{O}_Y$. This means g induces an isomorphism of ringed topological spaces $Y \simeq Z$.

Finally, we verify that $f = j \circ g$ as morphisms of ringed topological spaces. We already know this holds on the topological level. On the sheaf level, the map $f^{\#} : \mathcal{O}_X \to f_* \mathcal{O}_Y$ factors through $j^{\#} : \mathcal{O}_X \to \mathcal{O}_X / \mathcal{J} \simeq j_* \mathcal{O}_Z$ followed by the isomorphism $j_* \mathcal{O}_Z \simeq j_* g_* \mathcal{O}_Y = f_* \mathcal{O}_Y$, which is exactly $(j \circ g)^{\#}$.

Exercise 2.6

Let $f : X \to Y$, $g : Y \to Z$ be continuous maps, F a sheaf on X, and H a sheaf on Z. Show that

$$f^{-1}(g^{-1} H) = (g \circ f)^{-1} H,$$

$$g_*(f_* F) = (g \circ f)_* F.$$

Solution.

(a) The identity $f^{-1}(g^{-1}H) = (g \circ f)^{-1} H$.

Recall that for any sheaf $\mathcal{F}$ on a space Z and continuous map $\phi : W \to Z$, the inverse image sheaf $\phi^{-1} \mathcal{F}$ is defined as the sheaf associated to the presheaf

$$U \mapsto \varinjlim_{V \supseteq \phi(U)} \mathcal{F}(V).$$

For an open set $U \subseteq X$, we compute:

$$\begin{aligned} (f^{-1}(g^{-1}H))(U) &= \varinjlim_{V \supseteq f(U)} (g^{-1} H)(V) \\ &= \varinjlim_{V \supseteq f(U)} \left(\varinjlim_{W \supseteq g(V)} H(W) \right). \end{aligned}$$

On the other hand:

$$((g \circ f)^{-1} H)(U) = \varinjlim_{W \supseteq (g \circ f)(U)} H(W).$$

The key observation is that direct limits over directed sets satisfy:

$$\varinjlim_{V \supseteq f(U)} \varinjlim_{W \supseteq g(V)} H(W) = \varinjlim_{W \supseteq g(f(U))} H(W),$$

because every V containing $f(U)$ and W containing $g(V)$ also satisfies $W \supseteq g(f(U))$, and conversely any W containing $g(f(U))$ can be refined by some $g(V)$ with $V \supseteq f(U)$.

Intuition: The operations g^{-1} and f^{-1} both "restrict" functions to smaller and smaller neighborhoods. The order doesn't matter—we're ultimately restricting functions from neighborhoods of $(g \circ f)(U) = g(f(U))$.

(b) The identity $g_*(f_*F) = (g \circ f)_*F$.

For any open set $W \subseteq Z$, we compute:

$$\begin{aligned}(g_*(f_*F))(W) &= (f_*F)(g^{-1}(W)) \\ &= F(f^{-1}(g^{-1}(W))) \\ &= F((g \circ f)^{-1}(W)) \\ &= ((g \circ f)_*F)(W).\end{aligned}$$

Intuition: The pushforward f_* takes sections on X and "views them as sections on Y" via the map f. Applying g_* then views these as sections on Z. This is the same as directly viewing sections on X as sections on Z via $g \circ f$.

Exercise 2.13

Let $f : X \to Y$ be a continuous map of topological spaces. Let F be a sheaf on X and G a sheaf on Y. (a) Show that there exist canonical morphisms of sheaves

$$G \to f_*f^{-1}G,$$

$$f^{-1}f_*F \to F.$$

Moreover, if f is a closed immersion, then the first morphism is surjective. If f is an open immersion, then the second morphism is an isomorphism. (b) Using the maps of (2.1), show that there exists a canonical bijection

$$\mathrm{Mor}_Y(G, f_*F) \simeq \mathrm{Mor}_X(f^{-1}G, F).$$

Solution.

(a) The canonical morphisms.

First morphism: $G \to f_*f^{-1}G$.

For any open $V \subseteq Y$, we need a map $G(V) \to (f_*f^{-1}G)(V) = (f^{-1}G)(f^{-1}(V))$.

By definition of the inverse image:

$$(f^{-1}G)(f^{-1}(V)) = \varinjlim_{W \supseteq f(f^{-1}(V))} G(W).$$

Since $f(f^{-1}(V)) \subseteq V$, the set V is one of the neighborhoods over which we take the limit. Thus we have a canonical map

$$G(V) \hookrightarrow \varinjlim_{W \supseteq f(f^{-1}(V))} G(W) = (f_*f^{-1}G)(V)$$

coming from the inclusion of the indexing set.

Intuition: Sections of G over V can be "viewed as" sections over $f^{-1}(V)$ by restricting to smaller neighborhoods, then pushed forward to V.

Surjectivity when f is a closed immersion:

If f is a closed immersion, then f is a homeomorphism onto its image $f(X) \subseteq Y$. For any $x \in X$, the stalk map $G_{f(x)} \to (f_*f^{-1}G)_{f(x)}$ can be analyzed at the level of stalks. Since f identifies X with a closed subset of Y, the map becomes surjective because every section over neighborhoods of $f(x)$ can be pulled back to a section over neighborhoods of x and then pushed forward.

More concretely, using the adjunction property (part (b)), the map $G \to f_*f^{-1}G$ is the unit of the adjunction. When f is a closed immersion, the counit $f^{-1}f_*F \to F$ being an isomorphism (as f^{-1} is then exact) implies the unit is surjective.

Second morphism: $f^{-1}f_*F \to F$**.**

For any open $U \subseteq X$:

$$(f^{-1}f_*F)(U) = \varinjlim_{V \supseteq f(U)} (f_*F)(V) = \varinjlim_{V \supseteq f(U)} F(f^{-1}(V)).$$

Since $U \subseteq f^{-1}(V)$ whenever $V \supseteq f(U)$, we have restriction maps $F(f^{-1}(V)) \to F(U)$. These induce a map on the direct limit:

$$\varinjlim_{V \supseteq f(U)} F(f^{-1}(V)) \to F(U).$$

Intuition: A section of f_*F over V is a section of F over $f^{-1}(V)$. By taking the direct limit over neighborhoods V of $f(U)$ and restricting to U, we get a section over U.

Isomorphism when f is an open immersion:

If f is an open immersion, then $f : X \to f(X)$ is a homeomorphism and $f(X)$ is open in Y. For any open $U \subseteq X$, we can choose $V = f(U)$, which is open in Y and satisfies $f^{-1}(V) = U$. Then the direct limit is achieved by $V = f(U)$ itself, giving:

$$(f^{-1}f_*F)(U) = F(f^{-1}(f(U))) = F(U).$$

Thus the canonical map is an isomorphism.

(b) The adjunction bijection.

We need to construct a bijection:

$$\mathrm{Mor}_Y(G, f_*F) \simeq \mathrm{Mor}_X(f^{-1}G, F).$$

Construction. Given $\phi : G \to f_*F$, we define $\tilde{\phi} : f^{-1}G \to F$ as follows. For open $U \subseteq X$:

$$(f^{-1}G)(U) = \varinjlim_{V \supseteq f(U)} G(V).$$

The map $\phi_V : G(V) \to (f_*F)(V) = F(f^{-1}(V))$ exists for each V. Restricting to $U \subseteq f^{-1}(V)$ and passing to the direct limit gives:

$$\tilde{\phi}_U : (f^{-1}G)(U) \to F(U).$$

Inverse construction. Given $\psi : f^{-1}G \to F$, define $\bar{\psi} : G \to f_*F$ as follows. For open $V \subseteq Y$:

$$G(V) \to (f_*f^{-1}G)(V) = (f^{-1}G)(f^{-1}(V)) \xrightarrow{\psi_{f^{-1}(V)}} F(f^{-1}(V)) = (f_*F)(V).$$

The first map is the canonical morphism from part (a).

Naturality. Both constructions are natural in G and F, and checking that they are mutual inverses is a straightforward diagram chase using the universal property of direct limits.

Intuition: This is the **adjunction** between f^{-1} and f_*. The functor f^{-1} is *left adjoint* to f_*. In categorical terms:

$$\mathrm{Hom}(f^{-1}G, F) \cong \mathrm{Hom}(G, f_*F).$$

This is analogous to the situation in algebra where $- \otimes_A B$ is left adjoint to $\mathrm{Hom}_B(A, -)$, or in topology where the inverse image functor is left adjoint to the direct image functor.

Condition 3.2

Let us consider the following condition:

$$\begin{cases} X \text{ admits a covering by a finite number of affine} \\ \text{open subsets } \{U_i\}_i \text{ such that } U_i \cap U_j \text{ also admits} \\ \text{a finite covering by affine open subsets.} \end{cases} \tag{3.2}$$

Affine schemes always verify this condition, as do Noetherian schemes (Proposition 3.46(a)).

Proposition 3.12

Let X be a scheme and f OX(X). Then Xf is an open subset of X. Moreover, if X verifies condition (3.2) above, then the restriction OX(X) →OX(Xf) induces an isomorphism OX(X)f OX(Xf).

Definition 3.19

A closed subscheme of X is a closed subset Z of X endowed with the structure $(Z, \mathcal{O}_Z)$ of a scheme and with a closed immersion $(j, j^\#) : (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$, where $j : Z \to X$ is the canonical injection. In general, on a given closed subset Z, the structure of a closed subscheme of X is not unique (see, however, Proposition 4.2). It is therefore necessary always to specify the scheme structure that we want to put on Z.

Proposition 3.20

Let $X = \operatorname{Spec} A$ be an affine scheme. Let $j : Z \to X$ be a closed immersion of schemes. Then Z is affine and there exists a unique ideal J of A such that j induces an isomorphism from Z onto $\operatorname{Spec} A/J$.

Proof. Let us first show that Z verifies condition (3.2) above. There exist open subsets U_p of X such that $\{j^{-1}(U_p)\}_p$ is an affine open covering of Z. Each U_p is a union of principal open subsets $\{U_{pk}\}_k$ of X, and $j^{-1}(U_{pk})$ is affine because it is a principal open subset of $j^{-1}(U_p)$. By adding principal open sets covering $X \setminus j(Z)$, we obtain (since X is quasi-compact, see Exercise 1.6) a covering of X by a finite number of principal open sets $\{V_l\}_l$ such that $j^{-1}(V_l)$ is affine for every l. As $j^{-1}(V_l) \cap j^{-1}(V_{l'})$ is a principal open subset of $j^{-1}(V_l)$, and therefore affine, we indeed have condition (3.2).

Let $\mathcal{J} = \operatorname{Ker} j^\#$ and $J = \mathcal{J}(X)$. We know that Z is isomorphic to $V(\mathcal{J})$ endowed with the sheaf $\mathcal{O}_X/\mathcal{J}$ (Proposition 2.24). Let $g \in A$. We let h denote the image of g in $\mathcal{O}_Z(Z)$. From the exact sequence $0 \to \mathcal{J} \to A \to \mathcal{O}_Z(Z)$, we deduce an exact sequence by $\otimes_A A_g$:

$$0 \to \mathcal{J} \otimes_A A_g \to A_g = \mathcal{O}_X(D(g)) \to \mathcal{O}_Z(Z)_h.$$

Now $\mathcal{O}_Z(Z)_h = \mathcal{O}_Z(Z_h) = j_*\mathcal{O}_Z(D(g))$ (Proposition 3.12); therefore we have $\mathcal{J}(D(g)) = \mathcal{J} \otimes_A A_g$. Let $i : \operatorname{Spec} A/J \to X$ be the closed immersion defined in the preceding lemma. Then $(\operatorname{Ker} i^\#)(D(g)) = \mathcal{J}(D(g))$ for every principal open subset $D(g)$ of X. It follows that $\operatorname{Ker} i^\# = \mathcal{J}$. Hence $Z \simeq \operatorname{Spec} A/J$. Finally, the uniqueness of J comes from the fact that if we have an isomorphism $A/J \to A/J'$ which is compatible with the canonical surjections $A \to A/J$ and $A \to A/J'$, then $J = J'$.